

Comprehensive mapping of the exterior architecture of the dromedary camel brain.

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The morphological perspective of the camel brain remains largely unexplored. Therefore, studying the topography of the camel brain is of crucial importance. This study aimed to provide a detailed color-coded topographic representation of the camel brain's gross anatomy and nomenclature, showing its various gyri and sulci and their borders. We compared them to previously known information to develop a detailed description of camel brain exterior architecture. Our research identified distinctive gyri and sulci with discrete positions and surrounding structures, allowing us to define sulci boundaries and establish logical gyri nomenclature. This study uncovered previously overlooked gyri and sulci and improved descriptions of specific sulci. The ectomarginal sulcus, splenial sulcus, splenial gyrus, and ectogenual gyrus are a few examples. These findings highlight several unique anatomical features of the dromedary brain, which can guide future research. By providing a comprehensive examination of the distinctive exterior anatomical features of the camel brain, this study may serve as a point of convergence for all researchers, providing more accurate identification of the gyri and sulci.

Hierarchical functional differences between gyri and sulci at different scales.

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Gyri and sulci are 2 fundamental cortical folding patterns of the human brain. Recent studies have suggested that gyri and sulci may play different functional roles given their structural and functional heterogeneity. However, our understanding of the functional differences between gyri and sulci remains limited due to several factors. Firstly, previous studies have typically focused on either the spatial or temporal domain, neglecting the inherently spatiotemporal nature of brain functions. Secondly, analyses have often been restricted to either local or global scales, leaving the question of hierarchical functional differences unresolved. Lastly, there has been a lack of appropriate analytical tools for interpreting the hierarchical spatiotemporal features that could provide insights into these differences. To overcome these limitations, in this paper, we proposed a novel hierarchical interpretable autoencoder (HIAE) to explore the hierarchical functional difference between gyri and sulci. Central to our approach is its capability to extract hierarchical features via a deep convolutional autoencoder and then to map these features into an embedding vector using a carefully designed feature interpreter. This process transforms the features into interpretable spatiotemporal patterns, which are pivotal in investigating the functional disparities between gyri and sulci. We evaluate the proposed framework on Human Connectome Project task functional magnetic resonance imaging dataset. The experiments demonstrate that the HIAE model can effectively extract and interpret hierarchical spatiotemporal features that are neuroscientifically meaningful. The analyses based on the interpreted features suggest that gyri are more globally activated, whereas sulci are more locally activated, demonstrating a distinct transition in activation patterns as the scale shifts from local to global. Overall, our study provides novel insights into the brain's anatomy-function relationship.

Sex Differences in Insular Cortex Gyri Responses to the Valsalva Maneuver.

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Sex differences in autonomic regulation may underlie cardiovascular disease variations between females and males. One key autonomic brain region is the insular cortex, which typically consists of five main gyri in each hemisphere, and shows a topographical organization of autonomic function across those gyri. The present study aims to identify possible sex differences in organization of autonomic function in the insula. We studied brain functional magnetic resonance imaging (fMRI) responses to a series of four 18-s Valsalva maneuvers in 22 healthy females (age $\pm$ SD: 50.0 ± 7.9 years) and 36 healthy males (45.3 ± 9.2 years). Comparisons of heart rate (HR) and fMRI signals were performed with repeated measures ANOVA (threshold $P < 0.05$ for all findings). All subjects achieved the target 30 mmHg expiratory pressure for all challenges. Typical HR responses were elicited by the maneuver, including HR increases from ~ 4 s into the strain period (Phase II) and rapid declines to below baseline 5-10 s, following strain release (Phase IV). Small, but significant, sex differences in HR percent change occurred during the sympathetic-dominant Phase II (female male, i.e., greater undershoot in males). The insular cortices showed similar patterns in all gyri, with greater signal decreases in males than females. Both sexes exhibited an anterior-posterior topographical organization of insular responses during Phase II, with anterior gyri showing higher responses than more posterior gyri. The exception was the right anterior-most gyrus in females, which had lower responses than the four other right gyri. Responses were lateralized, with right-sided dominance during Phase II in both sexes, except the right anterior-most gyrus in females, which showed lower responses than the left. The findings confirm the anterior and right-sided sympathetic dominance of the insula. Although sex differences were prominent in response magnitude, organization differences between males and females were limited to the right anterior-most gyrus, which showed a lower fMRI response in females vs. males (and vs. other gyri in females). The sex differences suggest a possible differing baseline state of brain physiology or tonic functional activity between females and males, especially in the right anterior-most gyrus.

A functional model of cortical gyri and sulci.

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Diffusion tensor imaging (DTI) and high angular resolution diffusion imaging (HARDI) have been broadly used in the neuroimaging field to investigate the macro-scale fiber connection patterns in the cerebral cortex. Our recent analyses of DTI and HARDI data demonstrated that gyri are connected by denser, streamlined fibers than sulci are. Inspired by this finding and motivated by the fact that DTI-derived fibers provide the structural substrates for functional connectivity, we hypothesize that gyri are global functional connection centers and sulci are local functional units. To test this functional model of gyri and sulci, we examined the structural and functional connectivity among the landmarks on the selected gyral/sulcal areas in the frontal/parietal lobe and in the whole cerebral cortex via multimodal DTI and resting state fMRI (R-fMRI) datasets. Our results demonstrate that functional connectivity is strong among gyri, weak among sulci, and moderate between gyri and sulci. These results suggest that gyri are functional connection centers that exchange information among remote structurally connected gyri and neighboring sulci, while sulci communicate directly with their neighboring gyri and indirectly with other cortical regions through gyri. This functional model of gyri and sulci has been supported by a series of experiments, and provides novel perspectives on the functional architecture of the cerebral cortex.

Location of the central sulcus via cortical thickness of the precentral and postcentral gyri on MR.

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To determine whether relative cortical thickness measurements of the precentral and postcentral gyri can be used to differentiate the central sulcus from adjacent cortical sulci. Turbo inversion-recovery MR imaging of the entire brain was done with scans parallel to the anterior commissure-posterior commissure line. Cortical thickness was measured in each hemisphere with a jeweler's eyepiece with 0.1-mm gradations. Three measurements were obtained perpendicular to the central, precentral, and superior frontal sulci, as determined by means of established anatomic methods. The ratios of cortical thickness on both sides of the central, precentral, and superior frontal sulci were calculated and compared. The mean ratio of precentral/postcentral gyri was 1.64 for the right hemisphere and 1.53 for the left hemisphere. The mean cortical thickness ratios were as follows: 1.01 for the right hemisphere and 3.01 for the left hemisphere across the precentral sulcus, and 1.03 for the right hemisphere and 0.99 for the left hemisphere across the superior frontal sulcus. Cortical thickness measurements across the central sulcus provide a method for locating the primary motor (precentral gyri) and primary somatosensory (postcentral gyri) cortices. The higher mean cortical thickness ratio across the central sulcus corresponds with known cytoarchitectonic relationships.

The Fork sign: a new cortical landmark in the human brain.

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In the cerebral cortex, establishing the precise relationship between functional areas and the macroscopic anatomy of gyri and sulci has a paramount importance for the field of neuroimaging and neurosurgical interventions. The anatomical orientation should start with the identification of anatomical landmarks to set the anatomo-functional boundaries. The human central sulcus region stands out as a well-defined structural and functional unit housing the primary motor and sensory cortices and is considered as key region to be identified during brain surgery. While useful anatomical landmarks have been discovered, especially in the axial plane, the identification of this region in the sagittal plane remains sometimes difficult. Using cadaveric whole brains and multi-modal analysis of MRI brain scans, we systematically observed a tuning fork-shaped sulco-gyral configuration centred around the gyral continuum bridging the pre-central gyrus with the middle frontal gyrus. We provide evidence that this 'Fork sign' is a consistent morphological feature visible on the lateral surface of the brain and a reliable radioanatomical landmark for identifying central sulcus region structures on sagittal MRI images, including the motor hand area.

Differential Patterns of Gyral and Sulcal Morphological Changes During Normal Aging Process.

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The cerebral cortex is a highly convoluted structure with distinct morphologic features, namely the gyri and sulci, which are associated with the functional segregation or integration in the human brain. During the lifespan, the brain atrophy that is accompanied by cognitive decline is a well-accepted aging phenotype. However, the detailed patterns of cortical folding change during aging, especially the changing age-dependencies of gyri and sulci, which is essential to brain functioning, remain unclear. In this study, we investigated the morphology of the gyral and sulcal regions from pial and white matter

surfaces using MR imaging data of 417 healthy participants across adulthood to old age (21-92 years). To elucidate the age-related changes in the cortical pattern, we fitted cortical thickness and intrinsic curvature of gyri and sulci using the quadratic model to evaluate their age-dependencies during normal aging. Our findings show that comparing to gyri, the sulcal thinning is the most prominent pattern during the aging process, and the gyrification of pial and white matter surfaces were also affected differently, which implies the vulnerability of functional segregation during aging. Taken together, we propose a morphological model of aging that may provide a framework for understanding the mechanisms underlying gray matter degeneration.

Differential distributions of parvalbumin-positive interneurons in the sulci and gyri of the adult ferret cerebral cortex.

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Although accumulating evidence suggests that there are significant anatomical and histological differences between the sulci and gyri of the cerebral cortex, whether there is a difference in the distribution of interneurons between the two cortical regions remains largely unknown. In this study, we systematically compared the distributions of parvalbumin-positive interneurons among three neighboring gyrus and sulcus pairs-coronal gyrus and cruciate sulcus, anterior ectosylvian gyrus and rostral suprasylvian sulcus, and posterior ectosylvian gyrus and pseudosylvian sulcus-in the adult ferret cerebral cortex. We proposed a method to partition sulci and gyri into several specific subregions through the deepest points of the sulci and the highest points of gyri in the inner and outer cortical contours of coronal sections. We found that the density of parvalbumin-positive interneurons in the gyri was significantly higher than that in the sulci. Further study revealed that the density of PV interneurons in superficial cortical layers (layers 2/3 and layer 4) was comparable among the three pairs of sulci and gyri. However, the density of parvalbumin-positive interneurons in cortical layers 5/6 was significantly higher in gyri than in sulci. These results indicate that parvalbumin-positive interneurons are differently distributed in infragranular layers of cortical sulci and gyri.

Morphological patterns of the postcentral sulcus in the human brain.

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The morphological structure of the postcentral sulcus and its variability were investigated in 40 structural magnetic resonance images of the human brain registered to the Montreal Neurological Institute (MNI) proportional stereotaxic space. This analysis showed that the postcentral sulcus is not a single sulcus, but rather a complex of sulcal segments separated by gyri, which merge their banks at distinct locations. Most of these gyri are submerged deep within the sulcus and can be observed only by examining the depth of the sulcus, although a small proportion may be observed from the surface of the brain. In the majority of the examined cerebral hemispheres (73.75%), the postcentral sulcus is separated into two or three segments or, less frequently, into four or five segments (12.5%), or it remains continuous (13.75%). Examination of the in-depth relationship between the postcentral sulcus and the intraparietal sulcus revealed that these two sulci may appear to join on the surface of the brain but they are in fact always separated by a gyrus in the cortical depth. In 32.5% of the examined hemispheres, a dorsoventrally oriented sulcus, the transverse postcentral sulcus, is located anterior to the postcentral sulcus on the

lower part of the postcentral gyrus. Systematic examination of the morphology of the postcentral sulcus in the proportional stereotaxic space that is used in functional neuroimaging studies is the first step toward the establishment of anatomical-functional correlations in the anterior parietal lobe.

[Radioanatomy of the cerebral cortex. Practical guide of identification].

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Cerebral sulci identification process is based on several formal rules: it consists a series of steps allowing the identification of the major sulci. The process may be variable for each of us and we propose to read the images in certain order starting with the frontal lobe and its superior frontal gyrus than precentral and central gyri followed by the lateral fissure. Then the insular, lateral parietal and medial hemispheric regions. Finally the major sulci within the temporal and occipital lobes are described in the frontal plane. This presentation indicates the principal rules of cerebral lobes'identification, the basis of variability perception and the localization of small lesions allowing an accurate correlation with clinical findings and a suitable therapeutic procedure.
